

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA070 2

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4,BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

Code of Conduct:

I SHAIK KHATAM RIDWAN ¹⁹²⁵¹¹⁴⁴⁵ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:





Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and

stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized	Handles most cases; reasonable	Handles basic cases only	Incomplete or inefficient

Network Throughput and Packet Loss

- Total data received = 8000 bits
- Time = 4 seconds
- Packets sent = 200
- Packets received = 180
- Efficient if throughput > 1500 bps and packet loss $< 10\%$

Pseudocode:

Algorithm network performance

BEGIN

totalDataReceived = 8000

time = 4

Packets sent = 200

Packets Received = 180

IF time = 0 THEN

DISPLAY "Error: Time
cannot be zero."

STOP

END IF

IF throughput > 1500 AND
Packet loss percentage < 10 THEN
network status =
"Efficient"

ELSE
network status =
"Inefficient"

END IF

DISPLAY "Network
classification:", network status

END

Calculation

Throughput:

$$\begin{aligned}\text{Throughput} &= \text{Total Data Received} / \text{Time} \\ &= 8000 / 4 \\ &= 2000 \text{ bps}\end{aligned}$$

Packet loss percentage:

$$\begin{aligned}\text{Packet loss} &= (20 / 200) = 100 \\ &= 10\%\end{aligned}$$

Since throughput is 2000 bps, which is greater than 1500 bps, but packet loss is 10%, which is not less than 10%, the network is:

Network classification: Inefficient

2, Bandwidth Utilization Monitoring

Pseudocode:

Algorithm Monitor Bandwidth

BEGIN

DECLARE bandwidthusage []

DECLARE linkNames []

number of Links = 5

Link names [1] — "Link 1"

Link names [2] — "Link 2"

Link names [3] — "Link 3"

Link names [4] — "Link 4"

Link names [5] — "Link 5"

WHILE TRUE DO

DISPLAY

Link Names [i]. "is operating normally."

DISPLAY

"Utilization: "
bandwidth usage [i], "%"

END IF

END FOR

DISPLAY "monitoring cycle completed."

WAIT 10 seconds

END WHILE

END

Explanation

For every link:

- If utilization is 80% or below, the link is considered to be operating normally.
- If utilization is greater than 80%, the link is considered overloaded.

3, Best Communication Path Selection

The question asks for continuous monitoring of bandwidth, delay, and utilization.

Pseudocode:

ALGORITHM

Select Best Communication Path

BEGIN

DECLARE bandwidth[]

DECLARE delay[]

DECLARE utilization[]

DECLARE link status[]

DECLARE quality score[]

number of links - 6

WHILE TRUE DO

DISPLAY "..... Network

Link monitoring"

FOR i = 1 TO

number of links DO

Unavailable or over loaded."

DISPLAY "selecting

alternate path."

END IF

END FOR

DISPLAY "Network

monitoring cycle completed."

WAIT 10 seconds

END WHILE

END

Explanation of link quality Score

The algorithm considers three important factors:

- Higher band width $\rightarrow$ better
- Lower delay $\rightarrow$ better
- Lower utilization $\rightarrow$ better

The score used is:

$$\text{Quality score} = \frac{\text{Bandwidth}}{(\text{Delay} + 1)} \times (100 - \text{Utilization})$$